

Week 1 Writing Assignment (CS-EE5841)

Data Preprocessing and Perceptron

Total Points: 42

Instructions: Complete all **Tasks** for each question (Q1-Q3). Show all key steps using formulas and provide brief interpretation where needed.

Q1. Z-Score Outlier Detection (15 pts)

Question

Given dataset:

$$X = \{5, 3, 7, 8, 100\}$$

Mean: $\mu = 5.6$, Standard deviation: $\sigma = 3.0$

Tasks

- Compute the Z-score for $X = 100$
- Determine whether 100 is an outlier

Hint

$$Z = \frac{X - \mu}{\sigma}$$

Solution

— Write solutions for all tasks here

Interpretation: ..write some text here...

Q2. Min-Max Normalization (12 pts)

Question

Given dataset:

$$X = [10, 15, 20, 25, 30]$$

Tasks

- Apply Min-Max normalization to the dataset
- Compute normalized value for $X = 20$

Hint

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Solution

— Write solutions for all tasks here

Interpretation: ...write some text here...

Q3. Perceptron Weight Update (15 pts)

Question

Given:

$$x = [2, 1], \quad y = 1$$

Initial weights:

$$w = [0.5, -0.5], \quad b = 0$$

Learning rate:

$$\eta = 0.1$$

Activation:

$$\hat{y} = \text{step}(w \cdot x + b)$$

Tasks

- Compute predicted output $\hat{y}$
- Compute error
- Update weights and bias

Hint

$$\text{Error} = y - \hat{y}$$

$$w_{new} = w + \eta \cdot \text{error} \cdot x$$

$$b_{new} = b + \eta \cdot \text{error}$$

Solution

— Write solutions for all tasks here

Interpretation: ...write some text here...